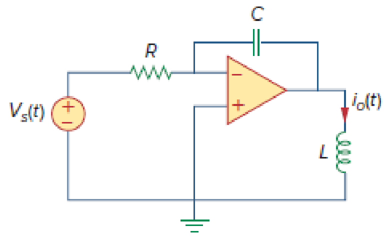
Chapter 16, Problem 81P

Problem

For the op-amp circuit in Fig. 16.99, find the transfer function, $T(s) = I(s)/Vs(s)$. Assume all initial conditions are zero.

Figure 16.99:



Step-by-step solution

Step 1 of 5

Refer to Figure 16.99 in the textbook.

Transform the circuit in Figure 16.99 to frequency domain.

The Laplace transform of $v_s(t)$ is $V_s(s)$ and $i_0(t)$ is $I(s)$.

The Laplace transform of inductance L is sL

The Laplace transform of capacitance C is $\frac{1}{sC}$

Assume that the all the initial conditions for the circuit should be zero.

Step 2 of 5

The frequency domain circuit is shown in Figure 1.

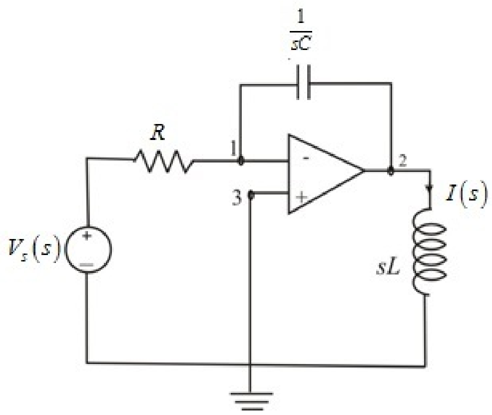


Figure 1

Step 3 of 5

Assume voltage at node 1 is $V_1(s)$, voltage at node 2 is $V_2(s)$, voltage at node 3 is $V_3(s)$.

Apply Kirchhoff's current law at node 1.

$$\frac{V_1(s)-V_s(s)}{R}+\frac{V_1(s)-V_2(s)}{1/sC}=0$$

$$V_1(s)\left[\frac{1}{R}+sC\right]-sCV_2(s)=V_s(s)\left[\frac{1}{R}\right] \quad \ldots\ldots (1)$$

$$V_1(s)\left[\frac{1}{R}+sC\right]-sCV_2(s)=V_s(s)\left[\frac{1}{R}\right]$$

Step 4 of 5

Use virtual ground concept, which is the voltage at positive terminal is equal to voltage at negative terminal, if negative feedback is provided in the circuit.

Therefore, voltage at node 3 is equal to voltage at node 1 and voltage at node 3 is equal to zero volts.

$$V_3(s)=0\ \text{V}$$

$$V_1(s)=V_3(s)$$

$$V_1(s)=0\ \text{V}$$

Substitute $0\ \text{V}$ for $V_1(s)$ in equation (1).

$$(0)\left[\frac{1}{R}+sC\right]-sCV_2(s)=V_s(s)\left[\frac{1}{R}\right]$$

$$V_s(s)\left[\frac{1}{R}\right]=-sCV_2(s)$$

$$V_2(s)=-\frac{V_s(s)}{sRC}$$

Step 5 of 5

The voltage at node 2 is calculated by using Ohm's law in Figure 1.

$$V_2(s)=sLI_o(s) \quad \ldots\ldots (2)$$

Substitute $-\frac{V_s(s)}{sRC}$ for $V_2(s)$ in equation (2).

$$sLI_0(s)=-\frac{V_s(s)}{sRC}$$

$$\frac{I_0(s)}{V_s(s)}=-\frac{1}{RLCs^2}$$

Therefore, the transfer function $T(s)$ is $\boxed{-\frac{1}{RLCs^2}}$.